

Database Management Systems — Test Document

A small synthetic DBMS study document for testing the LLM Context Optimizer.

1. Database Management System

A Database Management System (DBMS) is software used to define, create, store, retrieve, update, and manage data in databases. A DBMS provides controlled access to data and can enforce constraints, support transactions, provide security, and reduce unnecessary data redundancy.

Common DBMS concepts include tables, records, attributes, keys, relationships, constraints, transactions, indexing, normalization, and query processing.

2. Relational Model

The relational model represents data using relations, commonly implemented as tables. A row is a tuple and a column is an attribute. A relation has a schema describing its attributes and their domains. SQL is widely used to query and manipulate relational databases.

A primary key uniquely identifies each row in a relation. A foreign key references a key in another relation and can be used to represent relationships between tables.

3. Functional Dependencies

A functional dependency describes a constraint between attributes. For attributes X and Y , X determines Y , written $X \rightarrow Y$, when two tuples having the same value of X must also have the same value of Y . Functional dependencies are important in analyzing redundancy and normalization.

For example, if `StudentID` uniquely identifies a student, then $\text{StudentID} \rightarrow \text{StudentName}$. The determinant is the left-hand side of the dependency.

4. Normalization

Normalization is a database design technique used to organize attributes and relations to reduce redundancy and undesirable update anomalies. The main anomalies are insertion anomaly, deletion anomaly, and update anomaly.

First Normal Form (1NF) requires atomic attribute values and no repeating groups. Second Normal Form (2NF) requires 1NF and removal of partial dependencies of non-key attributes on a part of a composite candidate key. Third Normal Form (3NF) requires 2NF and removal of transitive dependencies of non-key attributes on candidate keys.

Boyce-Codd Normal Form (BCNF) is stronger than 3NF. A relation is in BCNF when every non-trivial functional dependency $X \rightarrow Y$ has X as a superkey.

5. Transactions

A transaction is a logical unit of database work. The ACID properties are Atomicity, Consistency, Isolation, and Durability. Atomicity means a transaction is treated as an all-or-nothing operation. Consistency means integrity constraints remain satisfied. Isolation controls the interaction between concurrent transactions. Durability means committed changes persist even after failures.

Concurrency control helps prevent problems such as lost updates, dirty reads, non-repeatable reads, and phantom reads. Locking protocols and timestamp-based methods are common approaches.

6. Indexing

An index is an auxiliary data structure that can speed up data retrieval. B+ trees are commonly used for database indexes because they support efficient search and range queries. Hash indexes can be useful for equality searches.

Indexes improve read performance but consume storage and can increase the cost of insert, update, and delete operations because the index may also need to be modified.

7. SQL

SQL provides commands for defining and manipulating relational data. SELECT retrieves data, INSERT adds rows, UPDATE modifies rows, and DELETE removes rows. JOIN operations combine related rows from multiple tables. GROUP BY can aggregate rows, while HAVING filters groups.

A database designer may also use views to represent reusable queries, stored procedures to encapsulate database operations, and triggers to execute actions automatically when specified database events occur.

8. Example Question Topics

A user asking questions about this document might ask: What is normalization? Explain 1NF, 2NF, and 3NF. What is a functional dependency? What are the ACID properties? How do indexes affect database performance? What is the difference between a primary key and a foreign key? These questions are useful for testing whether relevance scoring selects the appropriate chunks.